

Angular Architect Roadmap – 90 Days (1 Hour Daily)

Month 1 → Angular Core + Performance

Week 1: Change Detection & Lifecycle

- Day 1 → Change Detection (Default vs OnPush) → Implement OnPush in a component.
- Day 2 → Practice: markForCheck & detectChanges with ChangeDetectorRef.
- Day 3 → Learn: Zone.js → disable zones in one component and manually trigger CD.
- Day 4 → Review lifecycle hooks with console logs in all hooks.
- Day 5 → Apply: Optimize a list with OnPush + trackBy.
- Day 6 → Mini Project: Build a list app → measure perf difference (Default vs OnPush).
- Day 7 → Recap quiz: explain how Angular CD works.

Week 2: RxJS Core

- Day 8 → Learn map, filter, tap → Practice with HttpClient API.
- Day 9 → Learn higher-order mapping (switchMap, mergeMap, concatMap, exhaustMap).
- Day 10 → Error handling: catchError, retry, retryWhen.
- Day 11 → Memory leaks: use takeUntil, shareReplay.
- Day 12 → AsyncPipe vs manual subscription.
- Day 13 → Mini Project: Search box with debounce + switchMap.
- Day 14 → Recap: Draw RxJS operator decision tree.

Week 3: Signals + Standalone Components

- Day 15 → Intro to Signals → create a signal & bind to template.
- Day 16 → Derived & computed signals → practice with counters.
- Day 17 → Effects with signals → log side effects.
- Day 18 → Compare Signals vs RxJS.
- Day 19 → Standalone components intro.
- Day 20 → Convert existing app to standalone components.
- Day 21 → Review + practice with standalone + signals.

Week 4: Template & Performance Optimization

- Day 22 → Learn trackBy in *ngFor.
- Day 23 → Pure vs impure pipes.
- Day 24 → Optimize subscriptions with AsyncPipe.
- Day 25 → Preloading strategies for lazy modules.
- Day 26 → Analyze bundle size with --stats-json.
- Day 27 → Optimize by removing unused imports & modules.
- Day 28 → Mini Project: TODO app optimized with OnPush + signals.

Month 2 → State Management + Architecture

Week 5: Component Communication

- Day 29 → Inputs/Outputs basics.
- Day 30 → EventEmitter vs Services.
- Day 31 → Subject as event bus.
- Day 32 → Smart vs Dumb components.

- Day 33 → Facade pattern intro.
- Day 34 → Refactor app using facade pattern.
- Day 35 → Review communication methods.

Week 6: NgRx Fundamentals

- Day 36 → Setup NgRx Store.
- Day 37 → Actions + Reducers.
- Day 38 → Selectors.
- Day 39 → Effects.
- Day 40 → Entity adapter for lists.
- Day 41 → Mini Project: CRUD with NgRx.
- Day 42 → Review with NgRx DevTools.

Week 7: Component Store

- Day 43 → Setup ComponentStore.
- Day 44 → Selectors + updaters.
- Day 45 → Effects in ComponentStore.
- Day 46 → Compare with NgRx.
- Day 47 → When to use which.
- Day 48 → Mini Project: Feature module with ComponentStore.
- Day 49 → Review + refactor.

Week 8: Architecture & Patterns

- Day 50 → Nx Monorepo intro.
- Day 51 → Create Angular library with Nx.
- Day 52 → Dependency Inversion principle.
- Day 53 → Reusable services + interceptors.
- Day 54 → Facade pattern with NgRx.
- Day 55 → Refactor project with best practices.
- Day 56 → Review & write documentation.

Month 3 → Enterprise Features

Week 9: SSR & SEO

- Day 57 → Setup Angular Universal.
- Day 58 → CSR vs SSR rendering.
- Day 59 → Handle API calls with SSR.
- Day 60 → Static Site Generation.
- Day 61 → SEO optimizations with SSR.
- Day 62 → Mini Project: Blog with Angular Universal.
- Day 63 → Review.

Week 10: Testing

- Day 64 → Unit testing components.
- Day 65 → Testing services with HttpClientTestingModule.
- Day 66 → Component testing with TestBed.
- Day 67 → Mocking dependencies.
- Day 68 → Cypress/Playwright intro.

- Day 69 → E2E test for login flow.
- Day 70 → Review + coverage report.

Week 11: Security

- Day 71 → Route guards.
- Day 72 → Role-based access.
- Day 73 → JWT handling.
- Day 74 → XSS prevention.
- Day 75 → Security best practices.
- Day 76 → Mini Project: Secure Dashboard.
- Day 77 → Review.

Week 12: DevOps & Micro Frontends

- Day 78 → ESLint + Prettier setup.
- Day 79 → Strict mode advantages.
- Day 80 → GitHub Actions CI/CD.
- Day 81 → Dockerize Angular app.
- Day 82 → Monitoring with Sentry.
- Day 83 → Micro Frontends with Module Federation.
- Day 84 → Integrate Angular + React in MFE.
- Day 85 → Review.

Final Week: Capstone Project

- Day 86 → Plan project (E-commerce Dashboard).
- Day 87 → Setup Nx monorepo + NgRx.
- Day 88 → Add SSR + SEO.
- Day 89 → Add JWT auth + testing suite.
- Day 90 → Deploy + document architecture.